Comprehensive Algorithmic Compendium of the Dual-Branch Wavelet-ResNet Framework for Nuclear Instance Segmentation

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Abstract

This document presents the complete, exhaustive algorithmic specification of the Dual-Branch Wavelet-ResNet framework with Bi-Directional Channel Cross-Attention (BiCA), Learnable Sugeno Fuzzy Decision Fusion, and Marker-Controlled Distance-Guided Watershed Instance Decoding for histopathology nuclear instance segmentation. Every mathematical operation, tensor transformation, loss evaluation, sliding-window inference procedure, post-processing heuristic, validation search, and benchmarking metric is formulated into formal, self-contained, publication-grade pseudocode algorithms.

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1 Overview of Algorithmic Modules

The proposed histopathology segmentation system integrates 15 interconnected algorithmic procedures structured across six fundamental computational phases:

1. **Global Model Architecture**: End-to-end forward inference (Algorithm 1) and multi-scale bottleneck processing (Algorithm 5).
2. **Wavelet Multi-Resolution Processing**: Learnable Lifting Wavelet Transform forward decomposition (Algorithm 2) and symmetric lossless inverse synthesis (Algorithm 3).
3. **Feature Interactivity and Gating**: Generalized Mean (GeM) downsampling (Algorithm 4), Bi-Directional Channel Cross-Attention (Algorithm 6), and Skip-Connection Feature Alignment (Algorithm 7).
4. **Fuzzy Fusion and Supervision**: Learnable Sugeno Fuzzy Decision Fusion (Algorithm 8) and Multi-Task Composite Loss Evaluation (Algorithm 9).
5. **Optimization and Inference**: End-to-end model training with decoupled weight decay and EMA shadow parameters (Algorithm 10), followed by raised-cosine sliding-window tile stitching (Algorithm 11).
6. **Instance Decoding and Benchmarking**: Marker-controlled gradient watershed decoding (Algorithm 12), hyperparameter grid search (Algorithm 13), Aggregated Jaccard Index (Algorithm 14), and Panoptic Quality metric evaluation (Algorithm 15).

2 Global Architecture and Wavelet Lifting Formulations

Algorithm 1 DualBranchNet End-to-End Forward Pass

Require: Input RGB image batch $\mathbf{X} \in \mathbb{R}^{B\times 3\times H\times W}$
Ensure: Fused segmentation logits $\mathbf{z}_{\text{fused}} \in \mathbb{R}^{B\times 1\times H\times W}$,
Horizontal and vertical distance maps $\hat{\mathbf{H}}, \hat{\mathbf{V}} \in [-1,1]^{B\times 1\times H\times W}$,
Skip alignment loss $\mathcal{L}_{\text{feat}} \in \mathbb{R}$, Auxiliary logits $\mathbf{z}_{\text{aux3}}, \mathbf{z}_{\text{aux2}}$

// Stage 0: Learnable Stain Adaptation
$\mathbf{X}_{\text{adapt}} \leftarrow \text{StainAdapt}(\mathbf{X})$ ▷ Adaptive color normalization via 1×1 conv + residual

// Stage 1: Dual Encoder Processing
Initialize: Wavelet features $\mathbf{w}_0 \leftarrow \text{PReLU}(\text{BN}(\text{Conv}_{3\times 3}(\mathbf{X}_{\text{adapt}})))$ ▷ Spatial: $B\times 64\times H\times W$
Initialize: ResNet features $\mathbf{r}_0 \leftarrow \text{MaxPool}_{2\times 2}(\text{ReLU}(\text{BN}(\text{Conv1}_{7\times 7}(\mathbf{X}_{\text{adapt}}))))$ ▷ Spatial: $B\times 64\times \frac{H}{2}\times \frac{W}{2}$

for $l = 1$ **to** 4 **do**
  $\mathbf{w}_l, \mathbf{D}_l \leftarrow \text{LLWT_Stage}_l(\mathbf{w}_{l-1})$ ▷ Decompose via Algorithm 2; bypass details $\mathbf{D}_l$
  $\mathbf{r}_l \leftarrow \text{ResNet_Stage}_l(\mathbf{r}_{l-1})$ ▷ Standard residual bottleneck downsampling
end for

// Stage 2: Multi-Scale Context Bottleneck
$\mathbf{w}_4^{\text{aspp}} \leftarrow \text{ASPPBottleneck}(\mathbf{w}_4)$ ▷ Processed via Algorithm 5 with rates $\{1,4,7\}$
$\mathbf{r}_4^{\text{aspp}} \leftarrow \text{ASPPBottleneck}(\mathbf{r}_4)$

// Stage 3: Multi-Stage Decoders with Bi-Directional Cross-Attention
$\mathbf{d}_4^w \leftarrow \mathbf{w}_4^{\text{aspp}}, \quad \mathbf{d}_4^r \leftarrow \mathbf{r}_4^{\text{aspp}}$
$\mathcal{L}_{\text{feat}} \leftarrow 0.0$
for $l = 3$ **downto** 1 **do**
  // Upsample and Feature Extraction:
  $\mathbf{u}_l^w \leftarrow \text{ConvTranspose2d}(\mathbf{d}_{l+1}^w), \quad \mathbf{u}_l^r \leftarrow \text{ConvTranspose2d}(\mathbf{d}_{l+1}^r)$
  $\mathcal{L}_{\text{feat}} \mathrel{+}= \text{CosineDistLoss}(\mathbf{u}_l^r, \mathbf{r}_l)$ ▷ Evaluated via Algorithm 7

  // Bi-Directional Cross-Calibration:
  $\hat{\mathbf{u}}_l^w, \hat{\mathbf{u}}_l^r \leftarrow \text{BiCA}(\mathbf{u}_l^w, \mathbf{u}_l^r)$ ▷ Channel attention via Algorithm 6

  // Skip Connection Fusion and Convolutions:
  $\mathbf{d}_l^w \leftarrow \text{ConvBlock}\left([\hat{\mathbf{u}}_l^w \,\|\, \mathbf{w}_l]\right)$ ▷ Concatenation along channel dimension
  $\mathbf{d}_l^r \leftarrow \text{ConvBlock}\left([\hat{\mathbf{u}}_l^r \,\|\, \mathbf{r}_l]\right)$
end for

// Stage 4: Multi-Task Heads and Sugeno Fuzzy Fusion
$\mathbf{z}_w \leftarrow \text{Conv}_{1\times 1}(\mathbf{d}_1^w), \quad \mathbf{z}_r \leftarrow \text{Conv}_{1\times 1}(\mathbf{d}_1^r)$ ▷ Branch raw segmentation logits
$\mathbf{z}_{\text{fused}} \leftarrow \text{SugenoFusion}(\mathbf{z}_w, \mathbf{z}_r)$ ▷ Fuzzy integration via Algorithm 8

// Deep Supervision Auxiliary Heads:
$\mathbf{z}_{\text{aux3}} \leftarrow \text{BilinearUpsample}\left(\text{Conv}_{1\times 1}(\mathbf{d}_3^w + \mathbf{d}_3^r), \text{scale} = 4\right)$
$\mathbf{z}_{\text{aux2}} \leftarrow \text{BilinearUpsample}\left(\text{Conv}_{1\times 1}(\mathbf{d}_2^w + \mathbf{d}_2^r), \text{scale} = 2\right)$

// HoVer-Net Nuclear Distance Prediction Head:
$\mathbf{F}_{\text{hv}} \leftarrow \left[\mathbf{d}_1^w \,\|\, \mathbf{d}_1^r \,\|\, \text{BilinearUpsample}(\mathbf{d}_2^w + \mathbf{d}_2^r, 2)\right]$ ▷ Concatenated multi-scale features
$\hat{\mathbf{H}}, \hat{\mathbf{V}} \leftarrow \tanh\left(\text{Conv}_{1\times 1}\left(\text{ReLU}(\text{BN}(\text{Conv}_{3\times 3}^{4}(\mathbf{F}_{\text{hv}})))\right)\right)$ ▷ Distance maps in $[-1,1]$

return $\{\mathbf{z}_{\text{fused}}, \hat{\mathbf{H}}, \hat{\mathbf{V}}, \mathcal{L}_{\text{feat}}, \mathbf{z}_{\text{aux3}}, \mathbf{z}_{\text{aux2}}\}$

Algorithm 2 Polyphase Learnable Lifting Wavelet Forward Decomposition (LLWT)

Require: Feature tensor $\mathbf{X} \in \mathbb{R}^{B\times C\times H\times W}$

Ensure: Coarse approximation $c_l \in \mathbb{R}^{B\times C\times \frac{H}{2}\times \frac{W}{2}}$, Detail bypass $\mathbf{D} \in \mathbb{R}^{B\times 3C\times \frac{H}{2}\times \frac{W}{2}}$, Pooled output tensor $\mathbf{X}_{\text{pooled}} \in \mathbb{R}^{B\times 2C\times \frac{H}{2}\times \frac{W}{2}}$

// Step 1: Polyphase Sub-Grid Splitting

Extract four disjoint sub-lattices of shape $\mathbb{R}^{B\times C\times \frac{H}{2}\times \frac{W}{2}}$:

$x_{00}(b,c,i,j) \leftarrow \mathbf{X}(b,c,2i,2j)$ ▷ Even rows, even columns (reference coset)

$x_{01}(b,c,i,j) \leftarrow \mathbf{X}(b,c,2i,2j+1)$ ▷ Even rows, odd columns (horizontal phase)

$x_{10}(b,c,i,j) \leftarrow \mathbf{X}(b,c,2i+1,2j)$ ▷ Odd rows, even columns (vertical phase)

$x_{11}(b,c,i,j) \leftarrow \mathbf{X}(b,c,2i+1,2j+1)$ ▷ Odd rows, odd columns (diagonal phase)

// Step 2: Squeeze-and-Excitation Channel Attention Gating

$\mathbf{y}_{\text{gap}} \leftarrow \frac{1}{\frac{H}{2}\frac{W}{2}} \sum_{i=0}^{\frac{H}{2}-1} \sum_{j=0}^{\frac{W}{2}-1} x_{00}(:,:,i,j) \in \mathbb{R}^{B\times C\times 1\times 1}$

for $k \in \{h, v, d\}$ **do**

$\mathbf{s}_k \leftarrow \sigma\Big(\mathbf{W}_{2,k} \cdot \text{ReLU}\left(\mathbf{W}_{1,k}\mathbf{y}_{\text{gap}} + \mathbf{b}_{1,k}\right) + \mathbf{b}_{2,k}\Big)$ ▷ Channel scaling vector

$\tilde{\mathcal{P}}_k(x_{00}) \leftarrow \text{DWConv}_{3\times 3}^{(k)}(x_{00}) \odot \mathbf{s}_k$ ▷ Gated directional predictor

end for

// Step 3: Directional Detail Prediction (Dual Lifting Step)

$d_h \leftarrow x_{01} - \tilde{\mathcal{P}}_h(x_{00})$ ▷ Horizontal high-frequency detail

$d_v \leftarrow x_{10} - \tilde{\mathcal{P}}_v(x_{00})$ ▷ Vertical high-frequency detail

$d_d \leftarrow x_{11} - \tilde{\mathcal{P}}_d(x_{00})$ ▷ Diagonal high-frequency detail

// Step 4: Coarse Approximation Update (Primal Lifting Step)

$c_l \leftarrow x_{00} + k_h\mathcal{U}_h(d_h) + k_v\mathcal{U}_v(d_v) + k_d\mathcal{U}_d(d_d)$ ▷ k_h, k_v, k_d are learnable scalars

// Step 5: Subband Concatenation and Dimension Projection

$\mathbf{D} \leftarrow [d_h \,\|\, d_v \,\|\, d_d] \in \mathbb{R}^{B\times 3C\times \frac{H}{2}\times \frac{W}{2}}$ ▷ Preserved detail sub-bands for bypass

$\mathbf{X}_{\text{subbands}} \leftarrow [c_l \,\|\, \mathbf{D}] \in \mathbb{R}^{B\times 4C\times \frac{H}{2}\times \frac{W}{2}}$ ▷ Full decomposed spectrum

$\mathbf{X}_{\text{pooled}} \leftarrow \text{ReLU}\left(\text{BN}\left(\text{Conv}_{1\times 1}(\mathbf{X}_{\text{subbands}})\right)\right) \in \mathbb{R}^{B\times 2C\times \frac{H}{2}\times \frac{W}{2}}$

return $\{c_l, \mathbf{D}, \mathbf{X}_{\text{pooled}}\}$

Algorithm 3 Inverse Learnable Lifting Wavelet Decoding (ILLWT)

Require: Coarse approximation $c_l \in \mathbb{R}^{B\times C\times \frac{H}{2}\times \frac{W}{2}}$, Detail bypass $\mathbf{D} = [d_h \,\|\, d_v \,\|\, d_d] \in \mathbb{R}^{B\times 3C\times \frac{H}{2}\times \frac{W}{2}}$

Ensure: Reconstructed spatial feature lattice $\hat{\mathbf{X}} \in \mathbb{R}^{B\times C\times H\times W}$

// Step 1: Invert Primal Lifting Update Step

$x_{00} \leftarrow c_l - \Big(k_h\mathcal{U}_h(d_h) + k_v\mathcal{U}_v(d_v) + k_d\mathcal{U}_d(d_d)\Big)$ ▷ Exact algebraic subtraction

// Step 2: Invert Dual Lifting Prediction Step

$\mathbf{y}_{\text{gap}} \leftarrow \text{GAP}(x_{00})$ ▷ Recompute channel gating vectors identically

for $k \in \{h, v, d\}$ **do**

  $\mathbf{s}_k \leftarrow \sigma\Big(\mathbf{W}_{2,k} \cdot \text{ReLU}\left(\mathbf{W}_{1,k}\mathbf{y}_{\text{gap}} + \mathbf{b}_{1,k}\right) + \mathbf{b}_{2,k}\Big)$

  $\tilde{\mathcal{P}}_k(x_{00}) \leftarrow \text{DWConv}^{(k)}_{3\times 3}(x_{00}) \odot \mathbf{s}_k$

end for

$x_{01} \leftarrow d_h + \tilde{\mathcal{P}}_h(x_{00})$ ▷ Recover horizontal sub-phase

$x_{10} \leftarrow d_v + \tilde{\mathcal{P}}_v(x_{00})$ ▷ Recover vertical sub-phase

$x_{11} \leftarrow d_d + \tilde{\mathcal{P}}_d(x_{00})$ ▷ Recover diagonal sub-phase

// Step 3: Polyphase Synthesis Interlacing

Allocate empty tensor $\hat{\mathbf{X}} \in \mathbb{R}^{B\times C\times H\times W}$

$\hat{\mathbf{X}}(b, c, 2i, 2j) \leftarrow x_{00}(b, c, i, j)$

$\hat{\mathbf{X}}(b, c, 2i, 2j+1) \leftarrow x_{01}(b, c, i, j)$

$\hat{\mathbf{X}}(b, c, 2i+1, 2j) \leftarrow x_{10}(b, c, i, j)$

$\hat{\mathbf{X}}(b, c, 2i+1, 2j+1) \leftarrow x_{11}(b, c, i, j)$

return $\hat{\mathbf{X}}$ ▷ Lossless spatial reconstruction: $\hat{\mathbf{X}} \equiv \mathbf{X}$

3 Context Aggregation, Cross-Attention, and Feature Alignment

Algorithm 4 Generalized Mean (GeM) Pooling Downsampling Block

Require: Feature tensor $\mathbf{X} \in \mathbb{R}^{B\times C\times H\times W}$, Learnable exponent parameter $p \in [1,\infty)$ (init $p = 3.0$),
Window size $K = 2$, Stride $S = 2$, Numerical clamp epsilon $\epsilon = 10^{-6}$
Ensure: Downsampled feature tensor $\mathbf{Y} \in \mathbb{R}^{B\times C\times \frac{H}{2}\times \frac{W}{2}}$

Clamp learnable exponent: $p_{\text{eff}} \leftarrow \max(p, 1.0)$ ▷ Enforce convexity condition
Pre-clamp activations: $\tilde{\mathbf{X}} \leftarrow \max(\mathbf{X}, \epsilon)$ ▷ Prevent singularity in power-law operation
$\mathbf{X}_{\text{pow}} \leftarrow \tilde{\mathbf{X}}^{p_{\text{eff}}}$ ▷ Element-wise exponentiation
$\mathbf{S} \leftarrow \text{AvgPool2D}(\mathbf{X}_{\text{pow}}, \text{kernel} = K, \text{stride} = S)$ ▷ Spatial average over local window Ω
$\mathbf{Y} \leftarrow \mathbf{S}^{1.0/p_{\text{eff}}}$ ▷ Element-wise inverse exponentiation

return Y

Algorithm 5 Atrous Spatial Pyramid Pooling (ASPP) Bottleneck with Rates $\{1, 4, 7\}$

Require: Bottleneck feature tensor $\mathbf{X} \in \mathbb{R}^{B\times C_{\text{in}}\times H\times W}$, Output channels C_{out}, Dropout probability $p_{\text{drop}} = 0.1$
Ensure: Context-enriched bottleneck tensor $\mathbf{X}_{\text{bot}} \in \mathbb{R}^{B\times C_{\text{out}}\times H\times W}$

// **Branch 0: Standard** 1×1 **pointwise projection**
$\mathbf{b}_0 \leftarrow \text{ReLU}\Big(\text{BN}\big(\text{Conv}_{1\times1}(\mathbf{X})\big)\Big) \in \mathbb{R}^{B\times C_{\text{out}}\times H\times W}$

// **Branches 1–3: Dilated depthwise-separable convolutions at rates** $r \in \{1, 4, 7\}$
for $r \in \{1, 4, 7\}$ **do**
  $\mathbf{t}_{\text{dw}} \leftarrow \text{BN}\Big(\text{DWConv}_{3\times3,\,\text{dilation}=r,\,\text{pad}=r}(\mathbf{X})\Big)$
  $\mathbf{b}_r \leftarrow \text{ReLU}\Big(\text{BN}\big(\text{Conv}_{1\times1}(\mathbf{t}_{\text{dw}})\big)\Big) \in \mathbb{R}^{B\times C_{\text{out}}\times H\times W}$
end for

// **Branch 4: Global context image pooling**
$\mathbf{g} \leftarrow \text{GAP}(\mathbf{X}) \in \mathbb{R}^{B\times C_{\text{in}}\times1\times1}$
$\mathbf{g}_{\text{proj}} \leftarrow \text{ReLU}\Big(\text{BN}\big(\text{Conv}_{1\times1}(\mathbf{g})\big)\Big) \in \mathbb{R}^{B\times C_{\text{out}}\times1\times1}$
$\mathbf{b}_{\text{pool}} \leftarrow \text{BilinearInterpolate}(\mathbf{g}_{\text{proj}}, \text{size} = (H, W)) \in \mathbb{R}^{B\times C_{\text{out}}\times H\times W}$

// **Channel Concatenation and Projection**
$\mathbf{B}_{\text{concat}} \leftarrow [\mathbf{b}_0 \,\|\, \mathbf{b}_1 \,\|\, \mathbf{b}_4 \,\|\, \mathbf{b}_7 \,\|\, \mathbf{b}_{\text{pool}}] \in \mathbb{R}^{B\times 5C_{\text{out}}\times H\times W}$
$\mathbf{X}_{\text{proj}} \leftarrow \text{ReLU}\Big(\text{BN}\big(\text{Conv}_{1\times1}(\mathbf{B}_{\text{concat}})\big)\Big) \in \mathbb{R}^{B\times C_{\text{out}}\times H\times W}$
$\mathbf{X}_{\text{bot}} \leftarrow \text{Dropout2D}(\mathbf{X}_{\text{proj}}, p = p_{\text{drop}})$

return $\mathbf{X}_{\text{bot}}$

Algorithm 6 Bi-Directional Channel Cross-Attention (BiCA)

Require: Wavelet features $\mathbf{x}_w \in \mathbb{R}^{B\times C\times H\times W}$, ResNet features $\mathbf{x}_r \in \mathbb{R}^{B\times C\times H\times W}$
Ensure: Calibrated features $\hat{\mathbf{x}}_w, \hat{\mathbf{x}}_r \in \mathbb{R}^{B\times C\times H\times W}$

function CrossAttentionStream($\mathbf{x}_{\text{query}}, \mathbf{x}_{\text{key_val}}, \gamma$)
  Set spatial token count: $N \leftarrow H \cdot W$
  $\mathbf{Q} \leftarrow \text{Reshape}\Big(\text{BN}\big(\text{Conv}_{1\times1}(\mathbf{x}_{\text{query}})\big),\ (B, C, N)\Big) \in \mathbb{R}^{B\times C\times N}$
  $\mathbf{K} \leftarrow \text{Reshape}\Big(\text{BN}\big(\text{Conv}_{1\times1}(\mathbf{x}_{\text{key_val}})\big),\ (B, C, N)\Big) \in \mathbb{R}^{B\times C\times N}$
  $\mathbf{V} \leftarrow \text{Reshape}\Big(\text{Conv}_{1\times1}(\mathbf{x}_{\text{key_val}}),\ (B, C, N)\Big) \in \mathbb{R}^{B\times C\times N}$

  // Scaled Channel-Wise Affinity Matrix:
  $\mathbf{A} \leftarrow \frac{1}{\sqrt{N}}\big(\mathbf{Q}\mathbf{K}^T\big) \in \mathbb{R}^{B\times C\times C}$ ▷ Dot product across all N spatial locations
  $\tilde{\mathbf{A}} \leftarrow \text{Softmax}(\mathbf{A},\ \text{dim} = -1) \in \mathbb{R}^{B\times C\times C}$ ▷ Normalizing rows over channels

  // Value Aggregation and Residual Projection:
  $\mathbf{O} \leftarrow \tilde{\mathbf{A}}\mathbf{V} \in \mathbb{R}^{B\times C\times N}$
  $\mathbf{O}_{\text{spatial}} \leftarrow \text{Reshape}(\mathbf{O},\ (B, C, H, W))$
  $\hat{\mathbf{x}} \leftarrow \mathbf{x}_{\text{query}} + \gamma \cdot \mathbf{W}_{\text{out}}(\mathbf{O}_{\text{spatial}})$ ▷ γ is initialized to 0.0 for stable warmup
  return $\hat{\mathbf{x}}$
end function

$\hat{\mathbf{x}}_w \leftarrow$ CrossAttentionStream($\mathbf{x}_w, \mathbf{x}_r, \gamma_w$) ▷ Wavelet calibrated by ResNet semantics
$\hat{\mathbf{x}}_r \leftarrow$ CrossAttentionStream($\mathbf{x}_r, \mathbf{x}_w, \gamma_r$) ▷ ResNet calibrated by Wavelet details

return $\{\hat{\mathbf{x}}_w, \hat{\mathbf{x}}_r\}$

Algorithm 7 Skip-Connection Feature Alignment Loss ($\mathcal{L}_{\text{feat}}$)

Require: Upsampled decoder feature stages $\{\mathbf{u}_l^r\}_{l=1}^4$, Encoder skip feature stages $\{\mathbf{s}_l^r\}_{l=1}^4$, Numerical stabilization epsilon $\epsilon = 10^{-6}$

Ensure: Scalar alignment loss $\mathcal{L}_{\text{feat}} \in \mathbb{R}$

$\mathcal{L}_{\text{feat}} \leftarrow 0.0$

for $l = 1$ **to** 4 **do**

  Extract spatial dimensions: $B, C_l, H_l, W_l \leftarrow \text{Shape}(\mathbf{u}_l^r)$

  // L_2 **Normalization along channel dimension** C_l**:**

  $\mathbf{n_u} \leftarrow \sqrt{\sum_{c=1}^{C_l} (\mathbf{u}_l^r(b,c,y,x))^2} + \epsilon \in \mathbb{R}^{B\times 1\times H_l\times W_l}$

  $\mathbf{n_s} \leftarrow \sqrt{\sum_{c=1}^{C_l} (\mathbf{s}_l^r(b,c,y,x))^2} + \epsilon \in \mathbb{R}^{B\times 1\times H_l\times W_l}$

  $\hat{\mathbf{u}}_l^r \leftarrow \mathbf{u}_l^r \,/\, \mathbf{n_u}, \quad \hat{\mathbf{s}}_l^r \leftarrow \mathbf{s}_l^r \,/\, \mathbf{n_s}$ ▷ Unit-norm representations on hypersphere

  // **Mean Squared Error on Unit Hypersphere:**

  $\mathcal{E}_l \leftarrow \frac{1}{B\cdot H_l\cdot W_l} \sum_{b,y,x} \sum_{c=1}^{C_l} \left(\hat{\mathbf{u}}_l^r(b,c,y,x) - \hat{\mathbf{s}}_l^r(b,c,y,x)\right)^2$

  $\mathcal{L}_{\text{feat}} \mathrel{+}= \mathcal{E}_l$ ▷ Accumulate across all skip levels

end for

return $\mathcal{L}_{\text{feat}}$ ▷ Mathematically equivalent to minimizing $\sum_l 2\big(1 - \cos(\theta_l)\big)$

4 Fuzzy Fusion and Multi-Task Loss Formulations

Algorithm 8 Learnable Sugeno Fuzzy Decision Fusion

Require: Raw branch logits $\mathbf{z}_w, \mathbf{z}_r \in \mathbb{R}^{B\times 1\times H\times W}$,
Unconstrained learnable logit parameters $\theta_1, \theta_2 \in \mathbb{R}$, Epsilon $\epsilon = 10^{-6}$
Ensure: Fused probability map $p_{\text{fused}} \in [0,1]^{B\times 1\times H\times W}$, Fused logits $\mathbf{z}_{\text{fused}} \in \mathbb{R}^{B\times 1\times H\times W}$

// **Map branch predictions and fuzzy densities to** $[0,1]$**:**
$p_1 \leftarrow \sigma(\mathbf{z}_w) = \frac{1}{1+e^{-\mathbf{z}_w}}, \quad p_2 \leftarrow \sigma(\mathbf{z}_r) = \frac{1}{1+e^{-\mathbf{z}_r}}$
$g_1 \leftarrow \sigma(\theta_1), \quad g_2 \leftarrow \sigma(\theta_2)$ ▷ Singleton fuzzy densities (confidence measures)

// **Compute Maximum Confidence Bound:**
$\mathbf{M} \leftarrow \mathbb{I}(p_1 \geq p_2)$ ▷ Binary mask indicating branch dominance
$g_{\max} \leftarrow \mathbf{M} \cdot g_1 + (1 - \mathbf{M}) \cdot g_2$ ▷ Spatially varying density of dominant stream

// **Closed-Form Sugeno Fuzzy Integral Evaluation:**
$h_{\max} \leftarrow \max(p_1, p_2)$ ▷ Upper envelope
$h_{\min} \leftarrow \min(p_1, p_2)$ ▷ Lower envelope
$p_{\text{fused}} \leftarrow \max\Big(\min(h_{\max}, g_{\max}),\, h_{\min}\Big)$

// **Numerically Stabilized Logit Inversion:**
$p_{\text{safe}} \leftarrow \text{Clamp}(p_{\text{fused}},\, \epsilon,\, 1.0 - \epsilon)$
$\mathbf{z}_{\text{fused}} \leftarrow \ln(p_{\text{safe}}) - \ln(1.0 - p_{\text{safe}})$

return $\{p_{\text{fused}}, \mathbf{z}_{\text{fused}}\}$

Algorithm 9 Multi-Task Composite Loss Evaluation

Require: Model outputs $\{\mathbf{z}_{\text{fused}}, \hat{\mathbf{H}}, \hat{\mathbf{V}}, \mathcal{L}_{\text{feat}}, \mathbf{z}_{\text{aux3}}, \mathbf{z}_{\text{aux2}}\}$, Ground truth $\{\mathbf{M}_{\text{gt}}, \mathbf{H}^*, \mathbf{V}^*\}$
Ensure: Total scalar loss $\mathcal{L}_{\text{total}} \in \mathbb{R}$, Loss dictionary $\mathcal{D}_{\text{loss}}$

function SegLoss($\mathbf{z}, \mathbf{y}$)
  $\mathbf{p} \leftarrow \sigma(\mathbf{z})$
  $TP \leftarrow \sum \mathbf{p} \cdot \mathbf{y}, \quad FP \leftarrow \sum \mathbf{p} \cdot (1 - \mathbf{y}), \quad FN \leftarrow \sum (1 - \mathbf{p}) \cdot \mathbf{y}$
  $TI \leftarrow \frac{TP+1.0}{TP+0.7FN+0.3FP+1.0}$ ▷ Tversky Index with $\alpha = 0.7, \beta = 0.3$
  $\mathcal{L}_{\text{FT}} \leftarrow (1.0 - TI)^{4/3}$ ▷ Focusing exponent $\gamma = 4/3$
  $\mathcal{L}_{\text{BCE}} \leftarrow \text{BCEWithLogitsLoss}(\mathbf{z}, \mathbf{y})$
  return $\mathcal{L}_{\text{FT}} + 0.4\mathcal{L}_{\text{BCE}}$
end function

// 1. Primary and Deep Supervision Segmentation Losses
$\mathcal{L}_{\text{seg}} \leftarrow \text{SegLoss}(\mathbf{z}_{\text{fused}}, \mathbf{M}_{\text{gt}})$ ▷ Primary fused segmentation loss
$\mathcal{L}_{\text{seg}} \mathrel{+}= 0.4 \cdot \text{SegLoss}(\mathbf{z}_{\text{aux3}}, \mathbf{M}_{\text{gt}}) + 0.2 \cdot \text{SegLoss}(\mathbf{z}_{\text{aux2}}, \mathbf{M}_{\text{gt}})$ ▷ Deep supervision

// 2. Nuclear Distance Regression and MSGE Losses
$\mathbf{K}_x \leftarrow \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}, \quad \mathbf{K}_y \leftarrow \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$ ▷ Sobel differentiation operators
$\mathcal{L}_{\text{mse}} \leftarrow \text{MSE}(\hat{\mathbf{H}}, \mathbf{H}^*) + \text{MSE}(\hat{\mathbf{V}}, \mathbf{V}^*)$
$\mathcal{L}_{\text{msge}}^H \leftarrow \frac{\sum_{\mathbf{M}_{\text{gt}}} \left((\hat{\mathbf{H}} * \mathbf{K}_x - \mathbf{H}^* * \mathbf{K}_x)^2 + (\hat{\mathbf{H}} * \mathbf{K}_y - \mathbf{H}^* * \mathbf{K}_y)^2 \right)}{\sum \mathbf{M}_{\text{gt}} + \epsilon}$
$\mathcal{L}_{\text{msge}}^V \leftarrow \frac{\sum_{\mathbf{M}_{\text{gt}}} \left((\hat{\mathbf{V}} * \mathbf{K}_x - \mathbf{V}^* * \mathbf{K}_x)^2 + (\hat{\mathbf{V}} * \mathbf{K}_y - \mathbf{V}^* * \mathbf{K}_y)^2 \right)}{\sum \mathbf{M}_{\text{gt}} + \epsilon}$
$\mathcal{L}_{\text{msge}} \leftarrow \mathcal{L}_{\text{msge}}^H + \mathcal{L}_{\text{msge}}^V$
$\mathcal{L}_{\text{hv}} \leftarrow \mathcal{L}_{\text{mse}} + \mathcal{L}_{\text{msge}}$

// 3. Boundary Consistency Loss
$\mathbf{P} \leftarrow \sigma(\mathbf{z}_{\text{fused}})$
$|\nabla_S \mathbf{P}| \leftarrow \sqrt{(\mathbf{P} * \mathbf{K}_x)^2 + (\mathbf{P} * \mathbf{K}_y)^2}$ ▷ Predicted edge magnitude
$|\nabla_S \mathbf{M}_{\text{gt}}| \leftarrow \sqrt{(\mathbf{M}_{\text{gt}} * \mathbf{K}_x)^2 + (\mathbf{M}_{\text{gt}} * \mathbf{K}_y)^2}$ ▷ Ground truth edge magnitude
$\mathcal{L}_{\text{bnd}} \leftarrow \frac{1}{BHW} \| |\nabla_S \mathbf{P}| - |\nabla_S \mathbf{M}_{\text{gt}}| \|_1$

// 4. Composite Loss Assembly
$\mathcal{L}_{\text{total}} \leftarrow \mathcal{L}_{\text{seg}} + 0.1\mathcal{L}_{\text{feat}} + 1.0\mathcal{L}_{\text{hv}} + 0.5\mathcal{L}_{\text{bnd}}$
return $\{\mathcal{L}_{\text{total}}, \mathcal{D}_{\text{loss}}\}$

5 Optimization Regimen and Sliding-Window Inference

Algorithm 10 End-to-End Training Loop with AdamW, EMA, and Cosine Annealing

Require: Dataset loader $\mathcal{D}_{\text{train}}$, Model $\mathcal{M}_{\boldsymbol{\theta}}$, Total epochs $E_{\max} = 100$, Initial lr $\eta_0 = 10^{-4}$
Ensure: Optimal shadow model weights $\boldsymbol{\theta}^*_{\text{EMA}}$

Initialize AdamW optimizer: $\boldsymbol{\theta}$, $\eta = \eta_0$, $\beta_1 = 0.9, \beta_2 = 0.999$, weight decay $\lambda_{\text{wd}} = 10^{-4}$
Initialize CosineAnnealingWarmRestarts: $T_0 = 10, T_{\text{mult}} = 2, \eta_{\min} = 10^{-6}$
Initialize shadow weights: $\boldsymbol{\theta}_{\text{EMA}} \leftarrow \text{Clone}(\boldsymbol{\theta})$
$t_{\text{step}} \leftarrow 0$

for epoch $= 1$ **to** $E_{\max}$ **do**
    $\mathcal{M}_{\boldsymbol{\theta}}.\,\text{train}()$
    for each mini-batch $(\mathbf{X}_{\text{batch}}, \mathbf{Y}_{\text{batch}})$ **in** $\mathcal{D}_{\text{train}}$ **do**
        $t_{\text{step}} \leftarrow t_{\text{step}} + 1$
        Optimizer zero gradients: $\nabla_{\boldsymbol{\theta}}\mathcal{L} \leftarrow \mathbf{0}$
        Forward pass: $\mathcal{L}_{\text{total}} \leftarrow \text{EvaluateLoss}(\mathcal{M}_{\boldsymbol{\theta}}(\mathbf{X}_{\text{batch}}), \mathbf{Y}_{\text{batch}})$ ▷ Via Algorithm 9
        Backward pass: Compute gradients $\mathbf{g} \leftarrow \nabla_{\boldsymbol{\theta}}\mathcal{L}_{\text{total}}$ via PyTorch AMP
        Gradient clipping: $\mathbf{g} \leftarrow \mathbf{g} \cdot \min\left(1.0, \frac{1.0}{\|\mathbf{g}\|_2 + \epsilon}\right)$ ▷ Clip norm to 1.0
        Optimizer step: Update $\boldsymbol{\theta}$ using AdamW update rule

        // Dynamic Warmup Exponential Moving Average:
        $\alpha_t \leftarrow \min\left(0.999, \frac{1+t_{\text{step}}}{10+t_{\text{step}}}\right)$
        $\boldsymbol{\theta}_{\text{EMA}} \leftarrow \alpha_t \boldsymbol{\theta}_{\text{EMA}} + (1.0 - \alpha_t)\boldsymbol{\theta}$
    end for
    Update learning rate: Scheduler . step()
end for

return $\boldsymbol{\theta}^*_{\text{EMA}}$

Algorithm 11 Seamless Full-Image Sliding-Window Inference via Raised-Cosine Blending

Require: Full-size histology image $\mathbf{I} \in \mathbb{R}^{3\times H_{\text{full}}\times W_{\text{full}}}$, Patch size $P = 256$, Stride $S = 192$ (overlap 64), Model $\mathcal{M}$

Ensure: Full probability map $\hat{\mathbf{P}} \in [0,1]^{H_{\text{full}}\times W_{\text{full}}}$, Full HoVer map $\hat{\mathbf{V}}_{\text{full}} \in [-1,1]^{2\times H_{\text{full}}\times W_{\text{full}}}$

// **1. Construct 2D Separable Raised-Cosine (Hanning) Weighting Matrix:**

$w(n) \leftarrow 0.5\left(1 - \cos\left(\frac{2\pi n}{P-1}\right)\right), \quad n \in \{0,\dots,P-1\}$

$\tilde{w} \leftarrow \max(w, 10^{-3})$ ▷ Prevent zero boundary values

$\mathbf{W}_{\text{2D}}(i,j) \leftarrow \tilde{w}(i)\cdot\tilde{w}(j) \in \mathbb{R}^{P\times P}$

// **2. Initialize Full Canvasses and Normalization Accumulator:**

$\mathbf{A}_{\text{prob}} \leftarrow \mathbf{0}_{H_{\text{full}}\times W_{\text{full}}}$ ▷ Probability accumulator

$\mathbf{A}_{\text{hv}} \leftarrow \mathbf{0}_{2\times H_{\text{full}}\times W_{\text{full}}}$ ▷ HoVer distance accumulator

$\mathbf{\Omega} \leftarrow \mathbf{0}_{H_{\text{full}}\times W_{\text{full}}}$ ▷ Window weight normalizer

Compute patch top-left grid coordinates: $\mathcal{C} \leftarrow \{(y_k, x_k) \mid y_k = k_y \cdot S,\ x_k = k_x \cdot S\}$

// **3. Tiled Inference and Weighted Splatting:**

for each coordinate $(y_k, x_k) \in \mathcal{C}$ **do**

  Extract tile: $\mathbf{X}_{\text{patch}} \leftarrow \mathbf{I}[:, y_k : y_k + P, x_k : x_k + P]$

  Forward evaluation: $p_{\text{patch}}, \mathbf{v}_{\text{patch}} \leftarrow \mathcal{M}(\mathbf{X}_{\text{patch}})$

  Splat probability: $\mathbf{A}_{\text{prob}}[y_k : y_k + P, x_k : x_k + P] \mathrel{+}= p_{\text{patch}} \odot \mathbf{W}_{\text{2D}}$

  Splat HoVer: $\mathbf{A}_{\text{hv}}[:, y_k : y_k + P, x_k : x_k + P] \mathrel{+}= \mathbf{v}_{\text{patch}} \odot \mathbf{W}_{\text{2D}}$

  Splat weights: $\mathbf{\Omega}[y_k : y_k + P, x_k : x_k + P] \mathrel{+}= \mathbf{W}_{\text{2D}}$

end for

// **4. Exact Partition of Unity Normalization:**

$\hat{\mathbf{P}} \leftarrow \mathbf{A}_{\text{prob}} \,/\, \max(\mathbf{\Omega}, 10^{-6})$

$\hat{\mathbf{V}}_{\text{full}} \leftarrow \mathbf{A}_{\text{hv}} \,/\, \max(\mathbf{\Omega}, 10^{-6})$

return $\{\hat{\mathbf{P}}, \hat{\mathbf{V}}_{\text{full}}\}$

6 Instance Decoding, Post-Processing, and Evaluation Metrics

Algorithm 12 Marker-Controlled Watershed Instance Decoding

Require: Full probability $\hat{\mathbf{P}}$, Full HoVer maps $\hat{\mathbf{V}}_{\text{full}} = [\hat{\mathbf{H}}, \hat{\mathbf{V}}]$, Seed threshold τ_{seed}, Erosion radius r_{erode},
Minimum nuclear area $A_{\min} = 10$ pixels
Ensure: Nuclear instance label map $\mathbf{Y}_{\text{inst}} \in \mathbb{N}^{H \times W}$

Foreground mask: $\mathbf{M}_{\text{bin}} \leftarrow \mathbb{I}(\hat{\mathbf{P}} \geq 0.5)$

// **1. Directional Gradient Energy Landscape:**
$\mathbf{G}_h \leftarrow \sqrt{(\hat{\mathbf{H}} * \mathbf{K}_x)^2 + (\hat{\mathbf{H}} * \mathbf{K}_y)^2}$ ▷ Horizontal map gradient magnitude
$\mathbf{G}_v \leftarrow \sqrt{(\hat{\mathbf{V}} * \mathbf{K}_x)^2 + (\hat{\mathbf{V}} * \mathbf{K}_y)^2}$ ▷ Vertical map gradient magnitude
$\mathbf{G} \leftarrow \max(\mathbf{G}_h, \mathbf{G}_v)$ ▷ Combined horizontal and vertical boundary energy
$\tilde{\mathbf{G}} \leftarrow \frac{\mathbf{G} - \min(\mathbf{G})}{\max(\mathbf{G}) - \min(\mathbf{G}) + \epsilon} \in [0, 1]$ ▷ Normalized energy surface

// **2. Marker Seed Extraction and Morphological Conditioning:**
$\mathbf{S}_{\text{seed}} \leftarrow (\tilde{\mathbf{G}} < \tau_{\text{seed}}) \wedge \mathbf{M}_{\text{bin}}$ ▷ Centroid seeds with low internal gradient
if $r_{\text{erode}} > 0$ **then**
    $\mathcal{B} \leftarrow \text{DiskStructuringElement}(r_{\text{erode}})$
    $\mathbf{S}_{\text{seed}} \leftarrow \mathbf{S}_{\text{seed}} \ominus \mathcal{B}$ ▷ Morphological erosion to detach touching nuclei
end if
$\mathbf{M}_{\text{markers}} \leftarrow \text{ConnectedComponents}(\mathbf{S}_{\text{seed}}, \text{connectivity} = 8)$

// **3. Topographical Watershed Flooding:**
if $\text{Max}(\mathbf{M}_{\text{markers}}) == 0$ **then**
    $\mathbf{Y}_{\text{inst}} \leftarrow \text{ConnectedComponents}(\mathbf{M}_{\text{bin}}, \text{connectivity} = 8)$ ▷ Fallback if no seeds survive
else
    $\mathbf{T}_{\text{surface}} \leftarrow (1.0 - \hat{\mathbf{P}}) + \tilde{\mathbf{G}}$ ▷ Inverted probability barrier + gradient elevation
    $\mathbf{Y}_{\text{inst}} \leftarrow \text{Watershed}(\text{image} = \mathbf{T}_{\text{surface}}, \text{markers} = \mathbf{M}_{\text{markers}}, \text{mask} = \mathbf{M}_{\text{bin}})$
end if

// **4. Area Filtering:**
for each label $k \in \text{Unique}(\mathbf{Y}_{\text{inst}}) \setminus \{0\}$ **do**
    if $\sum \mathbb{I}(\mathbf{Y}_{\text{inst}} == k) < A_{\min}$ **then**
        $\mathbf{Y}_{\text{inst}}[\mathbf{Y}_{\text{inst}} == k] \leftarrow 0$ ▷ Remove spurious cellular debris
    end if
end for

return $\mathbf{Y}_{\text{inst}}$

Algorithm 13 Validation Grid Search for Watershed Hyperparameters

Require: Validation set $\{(\mathbf{I}_k, \mathbf{Y}_k^{\text{gt}})\}_{k=1}^{V}$, Candidate thresholds $\mathcal{T} = \{0.3, 0.4, 0.5, 0.6\}$,
Candidate erosion radii $\mathcal{R} = \{0, 1, 2, 3\}$, Model $\mathcal{M}$
Ensure: Optimal parameters $(\tau^*_{\text{seed}}, r^*_{\text{erode}})$

Precompute stitched predictions: $\{(\hat{\mathbf{P}}_k, \hat{\mathbf{V}}_k)\}_{k=1}^{V} \leftarrow \text{InferenceOnDataset}(\{\mathbf{I}_k\}, \mathcal{M})$
BestAJI $\leftarrow -1.0$, $\quad (\tau^*, r^*) \leftarrow (0.4, 1)$

for each $\tau \in \mathcal{T}$ **do**
  for each $r \in \mathcal{R}$ **do**
    $\text{AJI}_{\text{sum}} \leftarrow 0.0$
    for $k = 1$ **to** V **do**
      $\mathbf{Y}_{\text{pred}} \leftarrow \text{WatershedDecode}(\hat{\mathbf{P}}_k, \hat{\mathbf{V}}_k, \tau, r)$ ▷ Via Algorithm 12
      AJI_{sum} += $\text{EvaluateAJI}(\mathbf{Y}_{\text{pred}}, \mathbf{Y}_k^{\text{gt}})$ ▷ Via Algorithm 14
    end for
    $\text{MeanAJI} \leftarrow \text{AJI}_{\text{sum}}/V$
    if MeanAJI $>$ BestAJI **then**
      BestAJI $\leftarrow$ MeanAJI, $\quad \tau^* \leftarrow \tau, \quad r^* \leftarrow r$
    end if
  end for
end for

return (τ^*, r^*)

Algorithm 14 Aggregated Jaccard Index (AJI) Evaluation

Require: Predicted instance map $\mathbf{Y}_{\text{pred}} \in \mathbb{N}^{H \times W}$, Ground truth instance map $\mathbf{Y}_{\text{gt}} \in \mathbb{N}^{H \times W}$
Ensure: Scalar metric AJI $\in [0, 1]$

Extract ground truth object IDs: $\mathcal{G} \leftarrow \text{Unique}(\mathbf{Y}_{\text{gt}}) \setminus \{0\}$
Extract predicted object IDs: $\mathcal{P} \leftarrow \text{Unique}(\mathbf{Y}_{\text{pred}}) \setminus \{0\}$

if $|\mathcal{G}| == 0 \wedge |\mathcal{P}| == 0$ **then return** 1.0 ▷ Empty image correctly classified
end if
if $|\mathcal{G}| == 0 \vee |\mathcal{P}| == 0$ **then return** 0.0 ▷ Total false positive or false negative
end if

IntersectionSum $\leftarrow 0.0$, UnionSum $\leftarrow 0.0$, $\mathcal{U}_{\text{used}} \leftarrow \emptyset$

for each ground truth index $g \in \mathcal{G}$ **do**
    $\mathbf{M}_g \leftarrow \mathbb{I}(\mathbf{Y}_{\text{gt}} == g)$
    Candidate predictions: $\mathcal{C} \leftarrow \text{Unique}(\mathbf{Y}_{\text{pred}}[\mathbf{M}_g]) \setminus \{0\}$
    MaxIoU $\leftarrow 0.0$, $p^* \leftarrow$ None, $I^* \leftarrow 0$, $U^* \leftarrow 0$

    for each candidate $p \in \mathcal{C}$ **do**
        $\mathbf{M}_p \leftarrow \mathbb{I}(\mathbf{Y}_{\text{pred}} == p)$
        $I \leftarrow \sum(\mathbf{M}_g \wedge \mathbf{M}_p)$, $U \leftarrow \sum(\mathbf{M}_g \vee \mathbf{M}_p)$
        IoU $\leftarrow I \,/\, U$
        if IoU $>$ MaxIoU **then**
            MaxIoU $\leftarrow$ IoU, $p^* \leftarrow p$, $I^* \leftarrow I$, $U^* \leftarrow U$
        end if
    end for

    if $p^* \neq$ None $\wedge\, p^* \notin \mathcal{U}_{\text{used}}$ **then**
        IntersectionSum += I^*, UnionSum += U^*
        $\mathcal{U}_{\text{used}} \leftarrow \mathcal{U}_{\text{used}} \cup \{p^*\}$
    else
        UnionSum += $\sum \mathbf{M}_g$ ▷ Unmatched ground truth penalty
    end if
end for

// **Unassigned Prediction Penalty:**
for each $p \in \mathcal{P} \setminus \mathcal{U}_{\text{used}}$ **do**
    UnionSum += $\sum \mathbb{I}(\mathbf{Y}_{\text{pred}} == p)$
end for

return AJI $\leftarrow \frac{\text{IntersectionSum}}{\text{UnionSum}}$

Algorithm 15 Panoptic Quality (PQ) Metric Computation

Require: Predicted instance map $\mathbf{Y}_{\text{pred}} \in \mathbb{N}^{H \times W}$, Ground truth instance map $\mathbf{Y}_{\text{gt}} \in \mathbb{N}^{H \times W}$, Threshold $\theta = 0.5$

Ensure: Panoptic Quality PQ, Segmentation Quality SQ, Recognition Quality RQ

Extract IDs: $\mathcal{G} \leftarrow \text{Unique}(\mathbf{Y}_{\text{gt}}) \setminus \{0\}, \quad \mathcal{P} \leftarrow \text{Unique}(\mathbf{Y}_{\text{pred}}) \setminus \{0\}$
if $|\mathcal{G}| == 0 \wedge |\mathcal{P}| == 0$ **then return** $\text{PQ} = 1.0, \text{SQ} = 1.0, \text{RQ} = 1.0$
end if
if $|\mathcal{G}| == 0 \vee |\mathcal{P}| == 0$ **then return** $\text{PQ} = 0.0, \text{SQ} = 0.0, \text{RQ} = 0.0$
end if

// **1. Find Unique Matches with** IoU > 0.5**:**
$\mathcal{S}_{\text{iou}} \leftarrow [], \quad \mathcal{U}_g \leftarrow \emptyset, \quad \mathcal{U}_p \leftarrow \emptyset$
for each $g \in \mathcal{G}$ **do**
    $\mathbf{M}_g \leftarrow \mathbb{I}(\mathbf{Y}_{\text{gt}} == g)$
    for each $p \in \text{Unique}(\mathbf{Y}_{\text{pred}}[\mathbf{M}_g]) \setminus \{0\}$ **do**
        $\mathbf{M}_p \leftarrow \mathbb{I}(\mathbf{Y}_{\text{pred}} == p)$
        $\text{IoU} \leftarrow \frac{\sum(\mathbf{M}_g \wedge \mathbf{M}_p)}{\sum(\mathbf{M}_g \vee \mathbf{M}_p)}$
        if $\text{IoU} > \theta$ **then**
            $\mathcal{S}_{\text{iou}}.\text{append}(\text{IoU})$
            $\mathcal{U}_g \leftarrow \mathcal{U}_g \cup \{g\}, \quad \mathcal{U}_p \leftarrow \mathcal{U}_p \cup \{p\}$
            break ▷ By uniqueness theorem, at most one prediction can satisfy IoU > 0.5
        end if
    end for
end for

// **2. True Positives, False Positives, and False Negatives:**
$TP \leftarrow |\mathcal{S}_{\text{iou}}|, \quad FP \leftarrow |\mathcal{P}| - TP, \quad FN \leftarrow |\mathcal{G}| - TP$

// **3. Quality Factor Computations:**
$$\text{SQ} \leftarrow \begin{cases} \frac{1}{TP} \sum_{s \in \mathcal{S}_{\text{iou}}} s & \text{if } TP > 0 \\ 0.0 & \text{otherwise} \end{cases}$$
$\text{RQ} \leftarrow \frac{TP}{TP + 0.5FP + 0.5FN}$
$\text{PQ} \leftarrow \text{SQ} \times \text{RQ}$

return $\{\text{PQ}, \text{SQ}, \text{RQ}\}$